CCEF-Net: Cultural Context-Aware Evidence Fusion Network

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***Abstract*—This report summarizes the completed foundation work for CCEF-Net, a proposed system for verifying Bengali/Banglish memes using textual, visual, cultural, and external evidence. The project has completed dataset auditing and cleaning, trained three text-classification baselines, and compared their performance. Multimodal dataset preparation, Bengali meme annotation, evidence retrieval, and the proposed CCEF-Net architecture remain future work.**

***Index Terms*—Fake news detection, Bengali NLP, BanglaBERT, XLM-RoBERTa, multimodal misinformation, meme verification, evidence-grounded classification**

I. Project Objective

CCEF-Net aims to classify Bengali/Banglish memes into four categories: Authentic, Misleading/Out-of-Context, Satire/Humor, and Insufficient Evidence. The proposed framework combines Bengali OCR text, visual features, cultural context, and external evidence from news articles and fact-checking sources. Unlike conventional fake-news classifiers, the system is intended to verify claim–image consistency and provide evidence-grounded explanations.

II. Environment Setup and Dataset Preparation

The project repository, Python environment, dependencies, GPU checks, Google Colab T4 setup, and Google Drive workspace have been prepared. The BanFakeNews dataset was audited, revealing 61,581 records in the original `All60Kdataset.csv`: 48,678 authentic and 12,903 fake samples. The binary dataset was selected because the accompanying train, validation, and test files use a different four-class label system.

Data cleaning produced banfakenewsBinaryClean.csv, containing 58,621 records: 48,677 authentic and 9,944 fake samples. Duplicate records, empty text, URLs, and mentions were removed; whitespace was normalized; and headlines and article content were combined into model inputs. The cleaned dataset contains no duplicate records or missing values.

III. Text Baseline Experiments

Three text-classification experiments were completed using the cleaned dataset: standard BanglaBERT, class-weighted BanglaBERT, and XLM-RoBERTa. Table I summarizes the reported results.

TABLE I
Baseline Performance Comparison

Model	Accuracy	Macro-F1	Fake Recall
BanglaBERT (Standard)	96.00%	92.26%	78.00%
BanglaBERT (Weighted)	95.84%	92.03%	78.75%
XLM-RoBERTa	95.79%	91.72%	75.67%

Standard BanglaBERT achieved the best overall performance, with 96.00% accuracy and 92.26% Macro-F1. Class weighting slightly improved fake-class recall from 78.00% to 78.75%, but marginally reduced accuracy and Macro-F1. XLM-RoBERTa performed below BanglaBERT across all three reported metrics in these experiments.

Despite the strong overall results, fake-class recall remains a limitation: the standard BanglaBERT model misses approximately 22% of fake examples at the reported operating point. These findings apply to the current text dataset and do not establish performance on Bengali memes or multimodal misinformation.

IV. Current Status and Remaining Work

Approximately 40

Next, a Fakeddit subset will support text–image workflow development, followed by collecting and annotating 500–1,000 Bengali memes. Remaining tasks include Bengali OCR evaluation, cultural-context extraction, evidence retrieval, and multimodal CCEF-Net development.

Final evaluation will compare text-only, image-only, fusion, and full-model approaches, with ablation studies measuring the contributions of cultural context and evidence retrieval. Explainability and paper preparation will follow.

V. Conclusion

The text-classification foundation of CCEF-Net has been established, with three baseline models evaluated. Standard BanglaBERT achieved the best overall results, although limited fake-class recall remains a challenge. The planned contributions—Bengali meme benchmarking, cultural-context-aware multimodal fusion, evidence verification, and grounded explanations—remain to be implemented and validated.